



BIT1123/BISE2093/DIT1113
BIT1123 OBJECT ORIENTED PROGRAMMING

ASSIGNMENT 1 – INDIVIDUAL
(20%)

PREPARED BY:

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1. Introduction

In the Object Oriented Programming course (BIT1123) I have worked through a series of Java programming tutorials which progressively built my knowledge and understanding of program logic and Object Oriented Programming.

The tutorials began from the basic concepts of java programming and then advanced to the concepts of classes, objects, encapsulation, inheritance, abstraction, polymorphism, file handling and graphical user interfaces.

I also had hands-on experience using GitHub and GitHub Codespaces to complete the tutorials. I learned how to structure my source code into various tutorial folders, and how to organize my programming work into one repository.

2. Knowledge Gained from Each Tutorial

Tutorial 1 – Java Classes, Objects and Methods

With the help of Tutorial 1 I was able to understand the basic structure of java program. This was my first time using a Student class, and learned how to make a Student object and how to use methods to do something.

This tutorial provided me with a useful base since classes and objects are fundamental to OOP. I also got a little more at ease with writing and executing a java program. The goal of this tutorial is to learn how to use objects and methods. This tutorial is to learn the usage of objects and methods.

The ideas of Tutorial 1 were continued in Tutorial 2. Practiced to make an object of Student class and used methods of another class.

In this tutorial, I was able to enhance my knowledge on having a program divided into multiple classes for java. I learnt that having related data and methods within a class can help in making a program more understandable and organized.

Tutorial 3– Classes, Inheritance and Object Relationships

Tutorials 3 provided more practical experience with classes and relationships between objects. This project includes Person, Student and Lecturer classes and a main program using these classes.

It's been a great tutorial to understand that there can be different classes of objects with the same properties. I also learned more about inheritance and the ability for Object-Oriented Programming to model real-world relationships between classes.

Tutorial 5 – Encapsulation

I learned about encapsulation more practically in tutorial 5. Student class includes private data (Student ID, name, CGPA and programme).

This information is controlled by getters and setters. From this tutorial I learned the importance of using protected data in a class and the need to have controlled methods on a class to access and modify the data.

This was a significant advance in my knowledge of OOP, as I could see the actual use of the concept of Encapsulation in the actual code written in java.

Tutorial 6 – Inheritance

I learnt about inheritance in Tutorial 6, relating to employee and lecturer. The lecturer class is an extension of the employee class, with additional details such as subject and department.

I found out how to utilize information and functionality in a parent class, which can be reused in a child class. I've also discovered that `super()` can be used to invoke the parent constructor.

I understand the concept of inheritance now as it had been explained in this tutorial as it reduces duplication and would help a more specific class in extending the existing class.

This tutorial covers abstraction and polymorphism. In this tutorial, you will learn about abstraction and polymorphism.

Tutorial 7

more experience was gained with Object-Oriented Programming by creating the Appliance class and its subclasses including WashingMachine and Microwave.

I found the tutorial useful in getting my head around the concept of using a general class to represent common characteristics, and individual classes to represent specific behaviour. This helped me to get a better grasp of abstraction and polymorphism.

This was helpful to me because I was able to see how OOP can be used to make a program more flexible and organized when there are commonalities between the different objects.

Proper use of collection and input and file handling in Python.

Tutorials 8-9

we presented a more practical program that manages tasks. The program will store tasks in an ArrayList and get input from the user using a Scanner.

This program lets the user add three tasks and prints a list of the tasks. It also supports files with text to store and manipulate data.

This tutorial helped me to understand how several data objects can be used with Java and how data can be held outside the program. It also got me acquainted with some of the java collections and I/O libraries.

Tutorial 10 – GUI Quiz Application

The 10th tutorial was not a terminal based tutorial, unlike the previous ones, but one to create a graphical quiz application.

There are 2 classes in the project: Questions and QuizBattleGUI. Develops an interactive quiz with the use of Java GUI components and event handling.

It was a tutorial that got me the understanding that Java is not only used to write programs that are run from the terminal, but also to write interactive graphical applications. It also provided me with experience in dealing with GUI elements and user interactions.

3. Challenges Encountered

A big hurdle I faced was having several different java tutorials within the same repository on Github. There are several tutorials with their own main(), and sometimes, it may be confusing which program to run.

Yet another difficulty was in grasping the concept of abstraction and polymorphism, and the difference between them and encapsulation and inheritance. At first these concepts might seem the same but using different java classes, I got a better grasp on the purpose.

Difficulties were also encountered while using various types of Java programs. For instance some tutorials were just terminal-based applications, and Tutorial 10 needed a GUI.

4. How I Overcame the Challenges

I have used these to make it through by working out the project structure before running a program carefully. I was taught how to find the right tutorial folder and the right class with the main() method for each tutorial.

Again with OOP concepts, I was learning by reading the code and seeing how the classes were connected to each other. The examples from Student, Lecturer, employee and Appliance classes helped to clarify concepts.

I also repeated the execution of the programs several times and verified the errors if they were not correct. This made me more patient and systematic to solve programming problems.

5. Improvements in My Java Programming Skills

From working through tutorials I have become more capable of creating classes, objects, methods, constructors and using various Java libraries.

I have also gained more skills in using collections, user input, files and graphical interfaces. Most importantly I have become more able to read code and understand interactions between classes.

The step-by-step development of Java, which included classes, inheritance, encapsulation, abstraction, file handling and GUI programming made me understand the language more deeply.

6. Understanding of Object-Oriented Programming Concepts

My knowledge of Object-Oriented Programming has become much better due to practical tasks.

Firstly I learned that encapsulation could be used to hide information about class members by making all variables private and using getter and setter methods.

Secondly I learned that with the help of inheritance a class could extend another class and inherit some of its properties. This feature could be seen in relation between employee and lecturer.

Finally I got a better understanding of such OOP concepts as abstraction and polymorphism on the basis of appliance example. It showed me how one could use general classes and allow specific classes to implement their own functionality.

7. Future Learning Plans

I intend to learn more about the Java programming language in the future through doing more programming problems and creating larger projects.

I want to get a better understanding of advanced OOP concepts and to write cleaner and more maintainable code. My other goal is to gain some more knowledge about Java GUI development and other Java libraries.

The other aim is to enhance my debugging skills and problem solving ability to be able to find and correct programming errors faster.

8. Conclusion

Overall, the tutorials were a beneficial learning experience in Java. The tutorials took me from a basic understanding of java programming to more advanced concepts of Object Oriented Programming.

I was exposed to hands-on experience with classes, objects, methods, encapsulation, inheritance, abstraction, polymorphism, collections, file handling and graphical user interfaces.

I also got to know how to structure my programming on GitHub and how to have multiple Java projects in one repository. Learned to be more confident and independent in writing, running and debugging java programs due to the challenges faced.

9. GitHub Repository URL

<https://github.com/abdulrheemothman57-rgb/othman-abdalrahim-H-A-202501010595>